

CSE330: Numerical Methods

Assignment 06

Total Marks: 25

1. Consider a set of four data points: $f(0) = 3$, $f(4) = -2$, $f(-1) = 2$, $f(1) = 1$. In the following, you are asked to find the **best fit polynomial of degree 2** by using the **Discrete square/Least square approximation** method as follows:

- (2 marks) From the given data, write down the matrices A, b and x.
- (3 marks) Evaluate $A^T A$ and $\det(A^T A)$.
- (2 marks) Compute the best-fit polynomial of degree 2.

2. Consider the coordinates: $(x, f(x)) = (0, 1), (0.5, 1.4), (1, 1.7), (1.5, 2)$. In the following, you are asked to construct the **best-fit linear polynomial** by using the **QR-decomposition** method as follows:

- (2 marks) Construct the matrices A, b and x.
- (3 marks) Evaluate the orthonormal vectors q_1 and q_2 , and construct the matrix Q.
- (2 marks) Compute the matrix R.
- (3 marks) Using Q and R, evaluate the matrix x, and hence compute the best-fit linear polynomial.

3. A function is given by $f(x) = e^{0.5x} + \sin x$ which is to be integrated on the interval $[0, 2]$.

- (2 marks) Evaluate the **exact integral I(f)**.
- (2 marks) Compute the numerical integral by using the **Newton-Cotes formula with $n = 2$** .
- (4 marks) Evaluate the numerical integral $C_{1,4}$ by using the **Composite Newton-Cotes** formula and find the percentage relative error.